

Ahmed-Rafik-El Mehdi BAAHMED

Strasbourg, France | abaahmed@cesi.fr | ahmedrafikelmehdibaahmed@gmail.com
LinkedIn: [ahmed-rafik-el-mehdi-baahmed](#) | GitHub: [EagleEye1107](#) | Portfolio: [eagleeye1107.github.io](#)

Ph.D. candidate in Computer Science specializing in computationally efficient Federated and Distributed Learning for heterogeneous IoT Edge–Cloud systems. Research focuses on resource-aware participation, personalized knowledge transfer, temporal learning under evolving data, federated split learning, and efficient autonomous Edge AI. Experienced in algorithm design and experimental validation on heterogeneous physical edge infrastructures using Raspberry Pi and NVIDIA Jetson devices. Research interests include communication-efficient distributed learning, streaming and temporal data, personalization, resource-aware optimization, and autonomous edge intelligence.

EDUCATION

Ph.D. in Computer Science / Informatique **Arts et Métiers (ENSAM Paris) / CESI LINEACT (UR 7527), France**
Dec. 2023 – Defense scheduled November 2026

- **Thesis:** *Computational Efficiency of Federated Learning Techniques on Edge-Cloud Infrastructures.*
- **Keywords:** Federated Learning, Edge Computing, Computational Efficiency, Edge--Cloud Infrastructures, Internet of Things, System and Learning Heterogeneity, Client Selection, Personalization, Knowledge Transfer, Edge Intelligence.

Master's Degree & State Engineering Diploma in Computer Systems Engineering **ESI Sidi Bel Abbes, Algeria**
Sept. 2018 – Jul. 2023

- **Master Thesis:** *Explainable Artificial Intelligence on Graph Neural Network-Based Network Intrusion Detection Systems.*
- **Engineering Thesis:** *E-GNNExplainer - Single-Instance Explanation of Edge-Classification GNNs for Network Intrusion Detection.*
- **Keywords:** Computer Systems Engineering, Graph Neural Networks, Explainable AI, Cybersecurity, Network Intrusion Detection, Deep Learning.

PROFESSIONAL EXPERIENCE

Ph.D. Researcher in Federated Edge Learning **CESI LINEACT UR 7527, Strasbourg, France**
Dec. 2023 – Nov. 2026

- Research on computationally efficient Federated Edge Learning across heterogeneous IoT Edge–Cloud infrastructures, addressing participant selection, personalized knowledge transfer, temporal learning, and resource-efficient edge deployment.
- Developed **FedCDRP**, a data- and resource-aware participant-selection framework for improving computation-to-performance under heterogeneous client conditions.
- Developed **HiFEL-OCKT**, **TempAtt-MTL**, **HiFedOCCA**, and **EucNorm²-PGD** for hierarchical, objective-congruent and personalized knowledge transfer across heterogeneous clients, models, and IoT ecosystems.
- Developed **TeRR-SAtt** and **AMGF**, complementary components of a momentum-guided federated split-distillation framework enabling resource-efficient temporal learning and edge intelligence.
- Designed and deployed a heterogeneous physical Edge–Cloud testbed comprising a coordinating server and **20 Raspberry Pi/NVIDIA Jetson edge devices**, with distributed orchestration, communication, monitoring, and computational profiling.
- Evaluated federated learning under non-IID, resource-heterogeneous, and **continuously evolving temporal data**, including smart-building and Industrial IoT scenarios.

Adjunct Lecturer in Computer Science

CESI Engineering School, Strasbourg, France

Sep. 2024 – Apr. 2026

- Taught Artificial Intelligence, Object-Oriented Design and Programming, Computer Networks, System Administration, Arithmetic & Cryptography, Programming Paradigms, and Python for Signal Processing courses.
- Designed practical sessions and supported students in applying software engineering, AI, and systems concepts.

Deep Learning Research Intern

INSA Lyon – LIRIS Laboratory, Villeurbanne, France

Feb. – Jul. 2023

- Implemented and evaluated Graph Neural Network models for Network Intrusion Detection Systems.
- Developed explanation methods for edge-classification GNNs to improve the transparency of cybersecurity predictions.
- Built experimental workflows for model training, evaluation, and analysis using Python tools for deep learning.
- Contributed to research work on trustworthy AI, graph learning, and network security analytics.

Network Engineer Intern

HASNAOUI Group of Companies, Sidi Bel Abbès, Algeria

Sept. 2021

- Assisted with network administration, infrastructure monitoring, and technical support tasks.
- Gained practical experience with enterprise networking, system configuration, and troubleshooting.

PUBLICATIONS & SCIENTIFIC CONTRIBUTIONS

Journal Articles

- Ahmed-Rafik-El Mehdi BAAHMED et al., “Synergistic data-resource participant selection for efficient Federated Edge Learning in IoT ecosystems,” **Future Generation Computer Systems**, 2026.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “HiFEL-OCKT: Hierarchical Federated Edge Learning with Objective Congruence and Multi-Level Knowledge Transfer for IoT Ecosystems,” **Internet of Things**, 2026.

Conference Papers

- Ahmed-Rafik BAAHMED et al., “Momentum-Guided Federated Split Distillation for Personalized Temporal Edge Intelligence,” **ECML PKDD 2026, Naples, Italy, 2026**.
- Ahmed-Rafik BAAHMED et al., “Computational Efficiency through Momentum-Guided Split Distillation: Rethinking Federated Learning for Edge Autonomy” **ECML PKDD 2026 PhD Forum, Naples, Italy, 2026**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Hyperparameter Impact on Computational Efficiency in Federated Edge Learning,” **IWCMC 2024, Ayia Napa, Cyprus, 2024**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Using Graph Neural Networks for the Detection and Explanation of Network Intrusions,” **ECML PKDD 2023, Turin, Italy, 2023**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Computer Vision and Deep Learning-Based Support System for Visually Impaired People in an Outdoor Environment,” **COC’22, 2022**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Conception d’une Station Météo Intelligente,” **COC’21, 2021**.

EARLIER SELECTED PROJECTS & LEADERSHIP

Alzheimer-AI — Deep Learning for Medical Image Analysis

Nov. 2022 – Feb. 2023

- Led a team developing a computer vision and deep learning-based system for early Alzheimer’s disease detection.
- Explored multiple model architectures and image-processing approaches for medical decision-support tasks.

- Managed the project workflow, technical implementation, and experimental validation.

ArtificialEye — Computer Vision Support System for Visually Impaired People

Mar. 2022 – Jul. 2022

- Led a team developing a deep learning and computer vision-based assistive system for visually impaired users in outdoor environments.
- Contributed to model development, system design, and scientific dissemination through conference publication/presentation.

Vice-President — AlphaBit Scientific Club

Jan. 2021 – Oct. 2021

- Coordinated scientific, technical, engineering, and training activities within the student community.
- Supported the organization of workshops, coding activities, and knowledge-sharing events.

SKILLS

Machine Learning & AI: Federated Learning, Split Learning, Edge AI, Distributed Machine Learning, Deep Learning, Explainable AI, Computer Vision, Time-Series Forecasting, Computational Intelligence paradigms.

Neural Network Architectures: CNNs, RNNs (LSTM, GRU, SimpleRNN), Attention Mechanisms, Reservoir Computing, Graph Neural Networks, Spiking Neural Networks, Deep Neural Networks, Lightweight Neural Models.

Programming & Development: Python, C/C++, JavaScript, SQL, Pascal, Git, GitHub, Docker.

Frameworks & Libraries: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib.

Distributed Systems & Parallel Computing: Edge-Fog-Cloud architectures, Kafka, MQTT, multiprocessing, multithreading, resource monitoring, experimental benchmarking.

IoT & Embedded Systems: Raspberry Pi, Arduino, Wireless Sensor Networks, LoRa, Sigfox, IoT cloud platforms.

Data & Analytics: Data mining/analytics, model evaluation, performance profiling, accuracy-latency-memory-CPU trade-offs analysis.

Professional Skills: Research communication, scientific writing, project coordination, teaching, teamwork, technical mentoring, scientific and technical supervision.

Research Interests: Efficient AI, Edge Intelligence, Federated Learning, Collaborative IoT, Trustworthy Distributed Learning, Temporal Intelligence, Efficient and Effective AI Deployment.

Languages: Arabic: Native | French: Fluent | English: Fluent